Quantize PDL Model (INT8)

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Setting

- Dataset: cityscapes
  - Image shape: [512, 1024]
  - Size:
    - Images set: 11GB
    - Labeled image set: 241MB
  - Calibration dataset: 50 images
  - Eval set: 500 images
- Weight: PDL_cityscapes_weight
- Metric : mIoU

Workflow

Steps

1. Load the pre-trained FP32 PDL model
2. Prepare calibration dataset
3. Apply Cross-Layer Equalization (CLE) for better quantization
4. Create a QuantizationSimModel (Post-Training Quantization)
5. Calibrate the quantized model
6. Compare FP32 vs INT8 outputs
7. Experiment on advanced quantize techniques.
8. Export the quantized model

Code

Implement: [vphong-bos/QuantizedPDL_v2]

Turtorial notebook: k QuantizedPDL

Evaluation report

Metric	FP32	INT8	Delta
mIoU	74.2232	71.5821	▼ **2.6411**

INT8 Model config (Best model i think)

- Calib size: 50
- config [file: fully_symmetric.jso]n
  - Param: Symmetric.
  - Attention: Symmetric.
  - Per-channel: True.
- Applied techniques:
  - Sequential MSE.
  - Conv + batch norm folding.
- ```
  !cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
      --calib_images /kaggle/input/datasets/ippapi/cityscrapes/kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
      --export_path /kaggle/working/quantized_model \
      --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
      --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json \
      --enable_custom_conv_bn_fold \
      --enable_seq_mse
  ```
- Exported qdq .onnx format:
  - ```
    {'mIoU': 72.8064775466919, 'IoU_per_class': array([97.75153 , 82.27364 , 91.22738 , 41.60513 , 52.61761 , 60.914387,
           63.780624, 75.517075, 91.268745, 58.23394 , 94.40633 , 79.02299 ,
           62.608795, 93.784515, 61.804836, 80.33838 , 65.82809 , 56.560253,
           73.77872 ], dtype=float32), 'FPS': 0.3043443524749376, 'Avg_Inference_Time_ms': 3285.751786974095}
    ```
- Exported qop .onnx format:
  - ```
    {'mIoU': 68.49730014801025, 'IoU_per_class': array([96.84175 , 77.57307 , 89.37341 , 28.553644, 51.54847 , 57.016045,
           58.87595 , 72.33881 , 89.732315, 56.564667, 92.76196 , 76.010605,
           57.24979 , 92.98381 , 58.76916 , 70.94229 , 59.354305, 43.181564,
           71.77714 ], dtype=float32), 'FPS': 0.2870188666434609, 'Avg_Inference_Time_ms': 3484.0915222559743}
    ```
- Exported .pkl format:
  - ```
    {'mIoU': 72.82845377922058, 'IoU_per_class': array([97.698524, 81.9601  , 91.225006, 41.453423, 52.87414 , 61.217762,
           63.932205, 75.56153 , 91.33193 , 58.305992, 94.382645, 79.10338 ,
           62.747063, 93.84333 , 62.118893, 80.25157 , 65.16189 , 56.64479 ,
           73.92642 ], dtype=float32), 'FPS': 4.6556722661518295, 'Avg_Inference_Time_ms': 214.7917514018991}
    ```

```
================ Compare FP32 vs Quantized ================
---- Accuracy ----
FP32  mIoU : 74.2232
INT8  mIoU : 72.8285
Drop       : -1.3947

---- Correlation ----
PCC        : 0.954526
```

There are workarounds implemented in these model:

- Torch JIT tracing:
  - Detail:
    - While tracing mode, **control flows** in python are treated as **constant**
    - Possible lead to wrong graph trace, wrong params map,...
  - Solutions:
    - Change model code by remove or change control flow.
  - Results: **Stabilizing graph trace** of model.
- AdaRound implementation:
  - Detail:
    - AdaRound normally hooks into standard PyTorch modules. Which requires:
      - . weights: layer's weights
      - forward (): layer's behaviors.
        - **Layer** must consist of a **single op**.
    - PDL implementation contains custom Conv with forward fused conv and other.
      - **Norm and activation** are **fused** with **conv** sometimes.
      - ```
        def forward(self, x: torch.Tensor) -> torch.Tensor:
            x = super().forward(x)

            if self.norm is not None:
                x = self.norm(x)

            if self.activation is not None:
                x = self.activation(x)

            return x
        ```
    - Led to AdaRound give up on optimizing model weights.
  - Solutions:
    - Wipe these layers out from AdaRound.
      - ```
        model.semantic_head.decoder.res5.project_conv.convs.0
        model.semantic_head.decoder.res5.project_conv.convs.4.1
        model.instance_head.decoder.res5.project_conv.convs.0
        model.instance_head.decoder.res5.project_conv.convs.4.1
        ```

        - mplement Sequential MSE instead.
    - Results: (mIoU)
        - Applied neither AdaRound nor Sequential MSE: **36.421**
        - AdaRound with some layers wiped out: **70.3142**
        - Sequential MSE: **72.8284**
- Onnx runtime graph optimization:
    - Detail:
        - QuantizeLinear and DequantizeLinear are called many times, adding overheads
            - QuantizeLinear: 332
            - DequantizeLinear: 417
        - Flag caused huge error: `ort.GraphOptimizationLevel.ORT_ENABLE_ALL`
    - Solutions:
        - Reduce graph optimize level by basic only.
    - Results:
        - Increased FPS: **0.22** → **0.28**
        - Reduce size: **239.88 MB** → **60.64 MB**

Model examines

- AIMET exported model:
    - DequantizeLinear: 481
    - QuantizeLinear: 392
    - Total nodes: 1676.
- ONNX graph optimized model:

```
=== Operator Counts ===
QuantizeLinear   : 221
DequantizeLinear : 510
Total nodes      : 1123

=== All Operator Counts ===
Add                        16
AveragePool                2
BatchNormalization         83
Concat                     6
Conv                       88
DequantizeLinear           510
MaxPool                    1
MemcpyFromHost             96
MemcpyToHost               9
Mul                        1
QuantizeLinear             221
Relu                       81
Resize                     9
```

```
=== Initializer / Weight Details ===
model.backbone.res2.0.conv2.norm.weight_q
  shape: (64,)
  dtype: int8
  min: 20.0
  max: 127.0
  mean: 82.59375
  preview(10): [111  86  78 114 124  91  70 122  94  74]

model.backbone.res4.5.conv3.weight_q
  shape: (1024, 256, 1, 1)
  dtype: int8
  min: -128.0
  max: 114.0
  mean: -3.656475067138672
  preview(10): [-28  10 -33 -22  28  -9   0  14  20  -8]
```

- Traced onnx runtime with:
  - Optimize flag: ALL.
    - FP32 ops + Q + DQ are converted into INT8 ops. Examples:
      - Conv + Q + DQ → QLinearConv
  - Optimize flag: Basic:
    - Unused initialize ops are wiped out.
    - FP32 ops stay the same.

```
=== Op Summary ===
Conv                 provider=CPUExecutionProvider      count= 825  total_us=93268036
QLinearConv          provider=CPUExecutionProvider      count=1375  total_us=15755710
DequantizeLinear     provider=CPUExecutionProvider      count=9025  total_us=5842594
BatchNormalization   provider=CPUExecutionProvider      count=2075  total_us=4185496
QuantizeLinear       provider=CPUExecutionProvider      count=3700  total_us=3150336
Transpose            provider=CPUExecutionProvider      count=2675  total_us=1676521
Resize               provider=CPUExecutionProvider      count= 225  total_us=1311712
QLinearAdd           provider=CPUExecutionProvider      count= 400  total_us=319955
Concat               provider=CPUExecutionProvider      count= 150  total_us=242654
Relu                 provider=CPUExecutionProvider      count=  75  total_us=204414
ReorderInput         provider=CPUExecutionProvider      count=  25  total_us=151431
MaxPool              provider=CPUExecutionProvider      count=  25  total_us=92863
ReorderOutput        provider=CPUExecutionProvider      count=  25  total_us=38770
QLinearAveragePool   provider=CPUExecutionProvider      count=  50  total_us=14716
Mul                  provider=CPUExecutionProvider      count=  25  total_us=9411

=== Op Summary ===
Conv                 provider=CPUExecutionProvider      count=2200  total_us=125917489
DequantizeLinear     provider=CPUExecutionProvider      count=8975  total_us=7515734
QuantizeLinear       provider=CPUExecutionProvider      count=3850  total_us=3418264
Resize               provider=CPUExecutionProvider      count= 225  total_us=1326668
Relu                 provider=CPUExecutionProvider      count=2025  total_us=1234123
Add                  provider=CPUExecutionProvider      count= 400  total_us=847912
AveragePool          provider=CPUExecutionProvider      count=  50  total_us=491019
Concat               provider=CPUExecutionProvider      count= 150  total_us=245626
MaxPool              provider=CPUExecutionProvider      count=  25  total_us=209054
Mul                  provider=CPUExecutionProvider      count=  25  total_us=9935
```

Quantization tuning results

- Calib size: 50
- config file: fully_symmetric.json
  - Param: Symmetric.
  - Attention: Symmetric.
  - Per-channel: True.

- Applied techniques:
  - Sequential MSE.
  - Conv + batch norm folding.
-
```
!cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
    --calib_images /kaggle/input/datasets/ippapi/cityscrapes/kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json \
    --enable_custom_conv_bn_fold \
    --enable_seq_mse
```

- Exported qdq .onnx format:
  -
```
{'mIoU': 72.8064775466919, 'IoU_per_class': array([97.75153 , 82.27364 , 91.22738 , 41.60513 , 52.61761 , 60.914387,
       63.780624, 75.517075, 91.268745, 58.23394 , 94.40633 , 79.02299 ,
       62.608795, 93.784515, 61.804836, 80.33838 , 65.82809 , 56.560253,
       73.77872 ], dtype=float32), 'FPS': 0.3043443524749376, 'Avg_Inference_Time_ms': 3285.751786974095}
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       58.87595 , 72.33881 , 89.732315, 56.564667, 92.76196 , 76.010605,
       57.24979 , 92.98381 , 58.76916 , 70.94229 , 59.354305, 43.181564,
       71.77714 ], dtype=float32), 'FPS': 0.2870188666434609, 'Avg_Inference_Time_ms': 3484.0915222559743}
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- Exported .pkl format:
  -
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{'mIoU': 72.82845377922058, 'IoU_per_class': array([97.698524, 81.9601  , 91.225006, 41.453423, 52.87414 , 61.217762,
       63.932205, 75.56153 , 91.33193 , 58.305992, 94.382645, 79.10338 ,
       62.747063, 93.84333 , 62.118893, 80.25157 , 65.16189 , 56.64479 ,
       73.92642 ], dtype=float32), 'FPS': 4.6556722661518295, 'Avg_Inference_Time_ms': 214.7917514018991}
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```
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---- Accuracy ----
FP32  mIoU : 74.2232
INT8  mIoU : 72.8285
Drop       : -1.3947

---- Correlation ----
PCC        : 0.954526
```

```
!cd /kaggle/working/QuantizedPDL_v2 && python run_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 250 \
    --export_path /kaggle/working/quantized_model \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_cle \
    --enable_bn_fold \
    --bias_correction \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json
```

```
{'mIoU': 69.15483474731445, 'IoU_per_class': array([96.42345 , 75.92103 , 90.43474 , 47.980732, 50.266445, 57.969
78 ,
       62.643932, 73.17586 , 90.08401 , 58.881836, 93.46539 , 73.8837  ,
       51.734673, 92.280594, 55.32219 , 73.826706, 47.529865, 51.600624,
       70.51631 ], dtype=float32), 'FPS': 3.0206960273311108, 'Avg_Inference_Time_ms': 331.04952995999884}
```

```
!cd /kaggle/working/QuantizedPDL_v2 && python run_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
```

```
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_cle \
    --enable_bn_fold \
    --enable_bn_reestimation \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json
```

```
{'mIoU': 69.50953602790833, 'IoU_per_class': array([96.55355 , 76.14466 , 90.117615, 50.170906, 50.207966, 57.620
71 ,
       62.173283, 73.046005, 89.55285 , 56.86559 , 93.27528 , 75.41225 ,
       57.197674, 91.43471 , 56.554222, 72.52131 , 48.50957 , 52.648632,
       70.674515], dtype=float32), 'FPS': 3.0346207256938644, 'Avg_Inference_Time_ms': 329.5304719740061}
```

```
!cd /kaggle/working/QuantizedPDL_v2 && python run_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_cle \
    --enable_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json
```

```
{'mIoU': 71.40286564826965, 'IoU_per_class': array([97.22554 , 79.02532 , 89.80291 , 37.776737, 53.992958, 58.790
623,
       63.307285, 75.03262 , 90.43207 , 59.154743, 94.16924 , 78.421135,
       61.124313, 93.49232 , 68.88586 , 72.60822 , 58.104855, 52.10972 ,
       73.198006], dtype=float32), 'FPS': 3.047347259255978, 'Avg_Inference_Time_ms': 328.15426498001216}
```

```
!cd /kaggle/working/QuantizedPDL_v2 && python run_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json
```

```
{'mIoU': 71.40286564826965, 'IoU_per_class': array([97.22554 , 79.02532 , 89.80291 , 37.776737, 53.992958, 58.790
623,
       63.307285, 75.03262 , 90.43207 , 59.154743, 94.16924 , 78.421135,
       61.124313, 93.49232 , 68.88586 , 72.60822 , 58.104855, 52.10972 ,
       73.198006], dtype=float32), 'FPS': 3.0223492629601862, 'Avg_Inference_Time_ms': 330.8684447079977}
```

```
!cd /kaggle/working/QuantizedPDL_v2 && python run_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_bn_fold \
    --enable_cle \
    --enable_custom_conv_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json
```

```
{'mIoU': 72.82845377922058, 'IoU_per_class': array([97.698524, 81.9601  , 91.225006, 41.453423, 52.87414 , 61.217
762,
       63.932205, 75.56153 , 91.33193 , 58.305992, 94.382645, 79.10338 ,
       62.747063, 93.84333 , 62.118893, 80.25157 , 65.16189 , 56.64479 ,
       73.92642 ], dtype=float32), 'FPS': 4.42372614536896, 'Avg_Inference_Time_ms': 226.053776192015}
```

normal:

```
!cd /kaggle/working/QuantizedPDL_v2 && python run_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --enable_seq_mse \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_custom_conv_bn_fold
```

```
{'mIoU': 73.37806224822998, 'IoU_per_class': array([97.79636 , 82.25411 , 91.446365, 44.190365, 53.777462, 61.339
676,
       65.07018 , 75.64025 , 91.546074, 58.506096, 94.36168 , 79.19565 ,
       62.523415, 93.96265 , 62.08321 , 81.591675, 67.080765, 57.66273 ,
       74.154495], dtype=float32), 'FPS': 6.034824566318657, 'Avg_Inference_Time_ms': 165.70489978800106}
```

full:

```
!cd /kaggle/working/QuantizedPDL_v2 && python run_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --enable_seq_mse \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_custom_conv_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json \
    --enable_bias_correction --bias_corr_empirical_only \
    --enable_bn_reestimation \
    --enable_cle \
    --enable_bn_fold
```

```
{'mIoU': 72.82845377922058, 'IoU_per_class': array([97.698524, 81.9601  , 91.225006, 41.453423, 52.87414 , 61.217
762,
       63.932205, 75.56153 , 91.33193 , 58.305992, 94.382645, 79.10338 ,
       62.747063, 93.84333 , 62.118893, 80.25157 , 65.16189 , 56.64479 ,
       73.92642 ], dtype=float32), 'FPS': 4.6861682228136425, 'Avg_Inference_Time_ms': 213.39396121797472}
```

```
!cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --enable_adaround \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_custom_conv_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json \
    --enable_bias_correction --bias_corr_empirical_only \
    --enable_bn_reestimation \
```

```
    --enable_cle \
    --enable_bn_fold
```

```
{'mIoU': 72.95270562171936, 'IoU_per_class': array([97.72177 , 82.01696 , 91.25249 , 41.9472  , 52.799343, 61.203
05 ,
       64.152885, 75.63178 , 91.35283 , 58.296204, 94.33756 , 79.118195,
       62.703377, 93.77799 , 62.074173, 80.45696 , 66.63867 , 56.80769 ,
       73.81226 ], dtype=float32), 'FPS': 4.709671331488775, 'Avg_Inference_Time_ms': 212.32904158598467}
```

- All conv asym:

```
!cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --enable_seq_mse \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_custom_conv_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/full_sym_conv_asym.json
```

```
{'mIoU': 72.94939756393433, 'IoU_per_class': array([97.73102 , 82.02355 , 91.242966, 42.142406, 51.485138, 60.803
72 ,
       64.48205 , 75.1402  , 91.25901 , 57.986813, 94.28383 , 78.85833 ,
       61.713142, 93.881744, 63.177574, 80.52631 , 68.96566 , 56.592487,
       73.74261 ], dtype=float32), 'FPS': 4.014138473418237, 'Avg_Inference_Time_ms': 249.1194577920105}
```

- All relu asym:

```
!cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --enable_seq_mse \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_custom_conv_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/full_sym_relu_asym.json
```

```
{'mIoU': 72.68186211585999, 'IoU_per_class': array([97.68727 , 81.93432 , 91.1312  , 39.607616, 52.500862, 61.190
445,
       63.686718, 75.42028 , 91.305664, 58.510197, 94.37745 , 79.059555,
       62.6184  , 93.88742 , 61.86865 , 79.89828 , 65.22125 , 57.021626,
       74.028175], dtype=float32), 'FPS': 4.445992730783752, 'Avg_Inference_Time_ms': 224.92164530006266}
```

- All concat asym:

```
!cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --enable_seq_mse \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
```

```
    --enable_custom_conv_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/full_sym_concat_asym.json
```

```
{'mIoU': 72.82845377922058, 'IoU_per_class': array([97.698524, 81.9601  , 91.225006, 41.453423, 52.87414 , 61.217762,
       63.932205, 75.56153 , 91.33193 , 58.305992, 94.382645, 79.10338 ,
       62.747063, 93.84333 , 62.118893, 80.25157 , 65.16189 , 56.64479 ,
       73.92642 ], dtype=float32), 'FPS': 4.645884407376974, 'Avg_Inference_Time_ms': 215.24427047994322}
```

- All add asym:

```
!cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
    --calib_images /kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --enable_seq_mse \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --enable_custom_conv_bn_fold \
    --config_file /kaggle/working/QuantizedPDL_v2/config/full_sym_add_asym.json
```

```
{'mIoU': 72.68638014793396, 'IoU_per_class': array([97.69418 , 81.950005, 91.16619 , 39.575584, 52.33288 , 61.238396,
       63.83972 , 75.45817 , 91.35853 , 58.853985, 94.3993  , 79.081635,
       62.422234, 93.90689 , 61.85146 , 80.12421 , 65.71362 , 56.1399  ,
       73.93433 ], dtype=float32), 'FPS': 4.600186094447436, 'Avg_Inference_Time_ms': 217.38251006997962}
```

- All conv relu asym:

```
!cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
    --calib_images /kaggle/input/datasets/ippapi/cityscrapes/kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --config_file /kaggle/working/QuantizedPDL_v2/config/ful_sym_conv_relu_asym.json \
    --enable_custom_conv_bn_fold
```

```
{'mIoU': 72.81966209411621, 'IoU_per_class': array([97.73329 , 82.18376 , 91.15178 , 39.209473, 52.000298, 60.81414 ,
       64.296234, 75.22283 , 91.24504 , 58.616512, 94.30745 , 78.73788 ,
       62.063686, 93.894485, 62.733376, 80.193344, 68.276146, 57.080376,
       73.81351 ], dtype=float32), 'FPS': 3.8647213289296416, 'Avg_Inference_Time_ms': 258.75086840399854}
```

```
!cd /kaggle/working/QuantizedPDL_v2 && python build_sim_quantized_pdl.py \
    --calib_images /kaggle/input/datasets/ippapi/cityscrapes/kaggle/working/cityscapes/leftImg8bit/train --num_calib 50 \
    --export_path /kaggle/working/quantized_model \
    --save_quant_checkpoint /kaggle/working/checkpoint/panoptic_deeplab_int8_state_dict.pkl \
    --config_file /kaggle/working/QuantizedPDL_v2/config/fully_symmetric.json \
```

```
    --enable_custom_conv_bn_fold
```

```
{'mIoU': 72.87205457687378, 'IoU_per_class': array([97.72675 , 82.04639 , 91.20323 , 41.29154 , 52.706482, 61.193012,
       64.04397 , 75.53302 , 91.3238  , 58.213764, 94.33952 , 79.00047 ,
       62.654655, 93.80686 , 61.65154 , 80.45052 , 66.69547 , 56.855244,
       73.83277 ], dtype=float32), 'FPS': 0.1625806444131364, 'Avg_Inference_Time_ms': 6150.793679098006}
```